

03 - REFRACTIVE INDEX - SPECTROMETER

Aim

Determination of the refractive index of the prism material using a spectrometer

Materials and Apparatus

Spectrometer, light source, prism, reading lens and bulb, and black cloth.

Theory

Least count of the spectrometer:

Size of one division of the main scale =

..... divisions of the main scale = divisions of the Vernier scale

1 division of the Vernier scale =

The least count =

Part (A)

Determination of the prism angle,

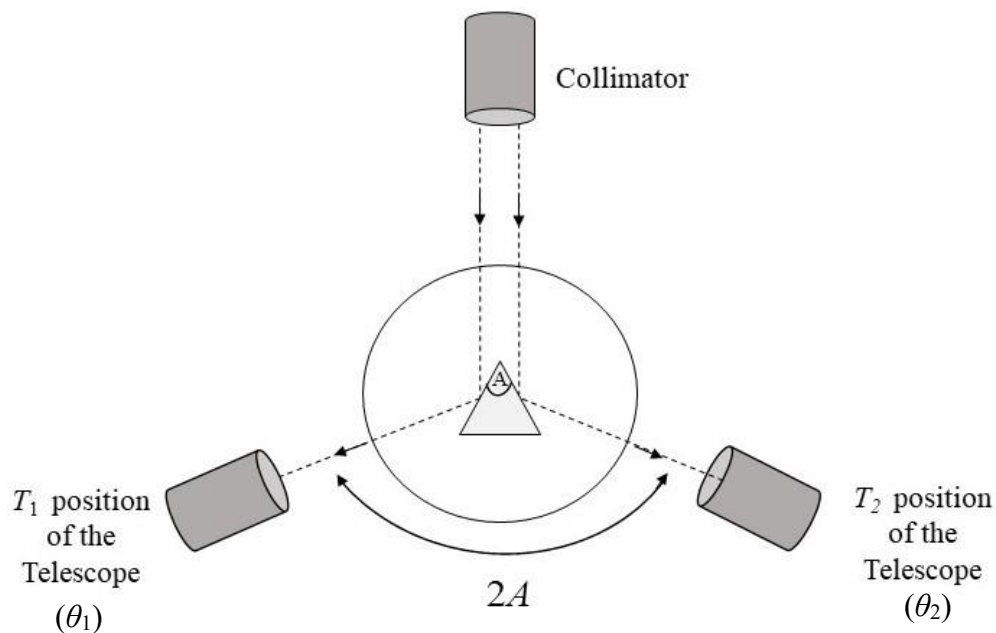


Fig. 1: Schematic diagram of the spectrometer arrangement

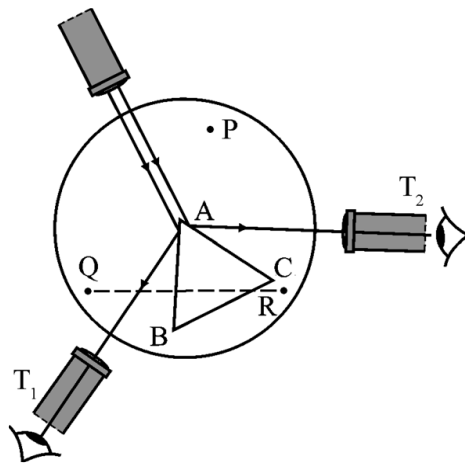


Fig. 2: Position of the prism during the leveling of the prism table

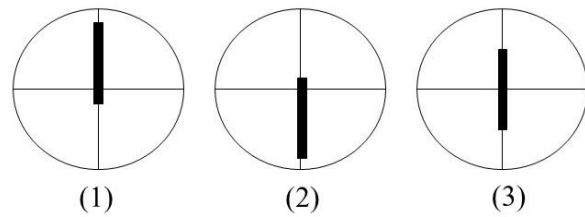


Fig. 3: Possible views of the slit before (1), (2) and after (3) adjustments

According to Figure 1,

$$\text{Prism angle } A = \frac{(\dots\dots\dots) - (\dots\dots\dots)}{2}$$

θ_1 and θ_2 are the readings of the spectrometer.

Part B

Determination of the minimum deviation point and refractive index of the prism material

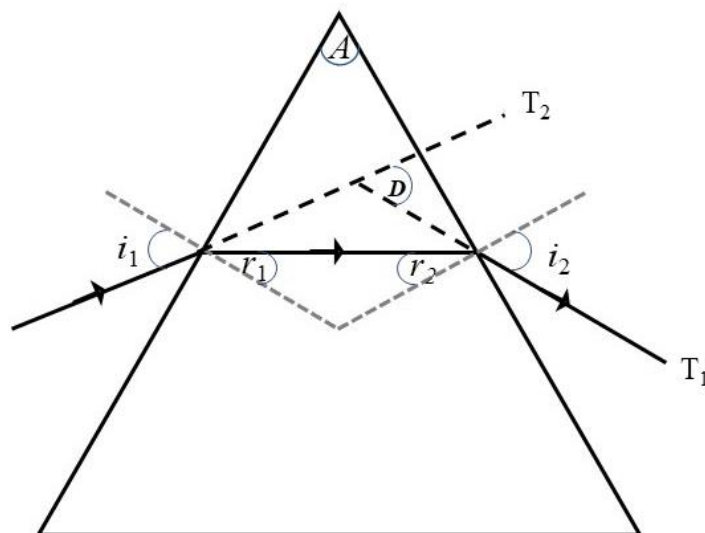


Fig. 3 : Ray diagram to show deviation by a prism

According to the Figure 3,

$$D = (i_1 - r_2) + (i_2 - r_2) \quad (1)$$

Also,

$$A = (r_1 + r_2) \quad (2)$$

From equation (1) and (2),

$$D = (i_1 + i_2) - A$$

As the incident angle i_1 changes, the angle of deviation, D also changes accordingly

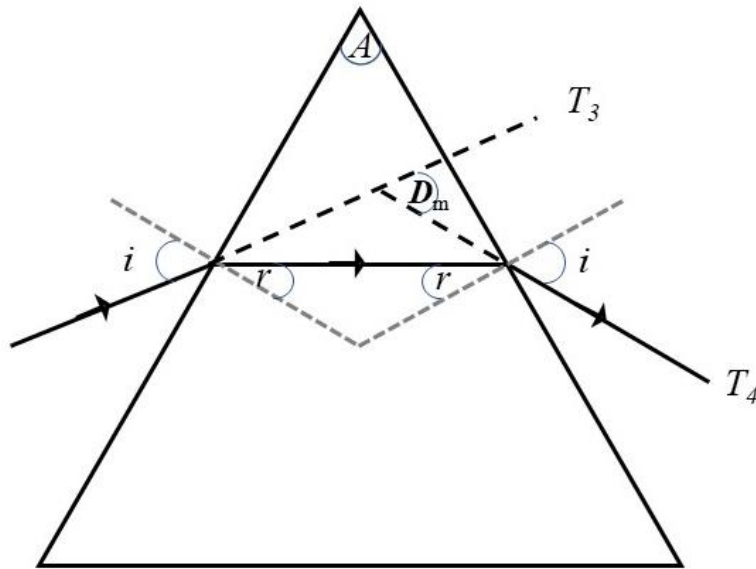


Fig. 4: Ray diagram to show minimum deviation by a prism

According to the figure 2, at the minimum deviation position

$$2r = A$$

$$2(i - r) = D_m$$

where,

i : angle of incidence

r : angle of reflection

D_m : minimum deviation angle ($T_4 - T_3$)

From Snell's law, the refractive index, η is given by

$$\eta = \frac{\sin i}{\sin r} = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Procedure

Part (A)

- a. Do the basic adjustments of the prism

Q₁ : What are the adjustments of the eye-piece?

Q₂ : How do you focus the telescope for this purpose?

Q₃ : How do you adjust the collimator for this purpose?

Q₄ : What is the purpose of placing the vertex of the prism close to the center of the prism table?

Q₅ : What is the purpose of placing a face (AB) perpendicular to the line (QR) joining two leveling screws?

Q₆ : Explain the selection of the screws of the prism table ?

Q₇ : Why do you have to repeat this process (moving between positions T_1 and T_2)?

- b. Get the readings from the spectrometer at both T_1 and T_2 positions

Part (B)

- a. Rotate the prism table to right by looking through the smooth surface until observe the refracted image of slit through it.

Q: How to identify the minimum deviation position?

- b. Bring the telescope to minimum deviation position
- c. Take the readings for the position of the telescope using both scales.
- d. Now remove the prism and rotate the telescope to the direct position and fix the telescope at this position.
- e. Take the readings for the direct positions using both scales.

Q: Is it possible to use a white light source for this experiment?

- f. Find D_m and calculate the refractive index of glass (η).